UNIVERSITY DEPT. OF COMPUTER ENGINEERING • BE SEMESTER VII CAPSTONE

Ansible-Provisioned Telecom Service Outage Dashboard

A Resilient, Containerized NOC Observability Platform with Git Flow, Continuous Testing, Jenkins CI/CD & Idempotent Infrastructure

Manas
Roll Number: 2
Class: **BE - VII (CMPN - C**
Academic Year

15 / 15	1.85h	188 MB	0 Drift	< 8
TESTS PASSED (100%)	OPTIMIZED MTTR	ALPINE JRE IMAGE	IDEMPOTENCY (CHANGED=0)	AUTOMA

⚠ 1. PROBLEM & TELECOM NEED

Telecom providers manage millions of active subscribers across 5G SA/NSA, 4G LTE, and DWDM fiber backbones. Incidents such as civic fiber excavations or baseband surges cause catastrophic regional drops in circles like Mumbai and Delhi.

Existing Bottlenecks: Siloed alarm systems, 45-min incident ingestion lag, manual SSH deployments prone to configuration drift, and lack of automated quality gates.

🏗 2. MULTI-TIER ARCHITECTURE

Tier 1: Presentation & Ingress
Nginx Reverse Proxy (Port 80) • Gzip • Forwarding to Spring Boot

Tier 2: Business Logic & REST
Java 17 • Spring Boot 3.2.5 • Embedded Tomcat 10.1 on Port 8085

Tier 3: CI/CD & Testing Quality Gate
Jenkins Pipeline as Code • Selenium E2E • JUnit 5 • Maven

Tier 4: Infrastructure & SRE
Docker Alpine Container • Ansible Playbooks • Systemd Unit

🛠 3. TECHNOLOGY STACK

Java 17 LTS · Spring Boot 3.2 · Thymeleaf 3 · Maven 3.9 · Selenium 4.18 · Headless Chrome · Jenkins CI/CD · Docker Alpine · Ansible 2.x · Nginx 1.24 · Systemd · Git Flow

💻 4. CORE DASHBOARD CAPABILITIES

1. **Data/Event Entry Modal:** Ingests incident title, circle, towers, affected subscribers, and root cause.
2. **Searchable Dashboard:** Real-time sub-second search and multi-attribute circle dropdown filtering.
3. **Summary Indicators:** Dynamic KPI telemetry for active outages, MTTR, and SLA adherence.
4. **Status Drill-Down:** Detailed modal showing cell towers, SLA countdown, and lifecycle state transitions.
5. **Critical Exception Alert View:** High-priority banner isolating active P1 carrier faults (≥25k users).

🔬 5. TESTING PYRAMID & SELENIUM

Automated test suite executing across all three pyramid tiers:

- **6 Unit Tests:** OutageService business logic & SLA calculations.
- **5 MockMvc Tests:** REST endpoint contract validation (200 OK).
- **4 Selenium E2E Journeys:** Navigation, Search, Ingestion, Drill-down in headless Chrome 155.
- **Failure Screenshot Watcher:** Automatically captures PNG on failure.

✓ 15 Tests Executed • 0 Failures • 0 Errors • 19.6s Total Time

⚙ 6. GIT FLOW & CONFLICT RESOLUTION

Implemented enterprise Git Flow with Pull Request peer reviews:

`feature/outage-reporting` → PR #1 merged to `develop`

`feature/alert-management` → Intentional conflict on `application.properties`

Harmonious resolution committed • Tagged `v1.0.0-mvp`

🐳 7. MULTI-STAGE DOCKE

Eliminates image bloat and eliminates attack

- **Stage 1 (Builder):** Maven 3.9 + JDK 17 wi caching to package the 25.4 MB executab
- **Stage 2 (Runtime):** Minimal Alpine JRE JAR.
- **Security Hardened:** Runs under unprivile
- **Result:** Image size reduced from 600 M **saving)**.

📡 8. ANSIBLE IAC & IDEM

Codified server provisioning across 16 discre

- Installs OpenJDK 17, Nginx, UFW, Curl.
- Renders Systemd supervisor unit & Nginx
- **Idempotency Proof:** Run 1: chang **changed=0**.
- **Automated Rollback:** `rollback.yml` res **seconds**.

📈 9. OPERATIONAL IMPAC

Operational Metric	Manual Legacy	O
Mean Time to Recovery	**4.2 hours**	1
Server Deployment	**45 mins (SSH)**	<
Rollback Duration	**30-60 mins**	<
Testing Coverage	**Manual sample**	1

Ansible-Provisioned Telecom Service Outage Dashboard • Student: **Manas Akhare** (Roll: 23102C0017) • Department of Computer Engineering HTTP Status: 200 UP • http://localhost